UNIVERSITY OF PERADENIYA

DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

END SEMESTER EXAMINATION - SEMESTER I (2021/2022)

CSC1041 - PROGRAMMING LABORATORY I

Answer **ALL** Questions Time Allowed: **Three** Hours

PART A - (C PROGRAMMING)

Use vi editor and save your C program as S20XXXQ1.c where S20XXX is your registration number. Extra marks will be awarded for using good programming practices.

1. [50 Marks]

You are required to create a C program for a Library Management System for the **librarians**. The librarian should be able to add any number of book records at a time (Hint: You may ask the librarian the number of records needed to add at a time).

a) A recurring menu should be implemented and when the add book details option is selected, the program should ask for book id, book name, author, number of copies and price for each book. The appearance of the main menu is as follows.

```
Example output:
              MENU
-------------------------------------
PRESS 1.TO ADD BOOK DETAILS.
PRESS 2.TO DISPLAY BOOK DETAILS.
PRESS 3.TO DISPLAY BOOK OF GIVEN AUTHOR.
PRESS 4.TO COUNT NUMBER OF BOOKS AND VALUE OF ALL THE BOOKS.
PRESS 5.FINE DETAILS.
PRESS 6.TO EXIT.
-------------------------------------
Enter Your Choice:
```

b) The details for all the books in the library, should be displayed as follows when the option 2 (Display Book Details) is selected.

```
Example output:
Enter Your Choice: 2
                Details of All Books
-----------------------------------------------------------
Book Id.        Book Name        Author Name     No. of Copies   Price
------------------------------------------------------------
 2001               Divergent       V.Rotch          3         1760.00
 4002               Insurgent       V.Rotch          5         1550.00
 1400               Cinder          M.Meyer          2         850.00
```

c) When option 3 is selected, librarians should be able to search for all the books written by a particular author.

```
Example output:
Enter Your Choice: 3
Enter Author Name: V.Rotch
--------------------------------------
 2001               Divergent       V.Rotch          3         1760.00
 4002               Insurgent       V.Rotch          5         1550.00
```

d) If the librarian selects option 4, the program should return the total book count in the library (including all the copies from the same book) and the total worth of the entire library assets. You may use the following equations.

Total book count = Σ (#of copies from each book)
Total worth of a book = Number of copies * Price
Total worth of the entire library = Σ (# of copies from a book * book price)

```
Example output:
-------------------------------------
Enter Your Choice: 4
Total Number of Books in Library : 10
Total worth of the entire library: 14730.00
```

e) A fine/penalty is charged from the members when not returning the borrowed books before the due date. The fine is calculated by considering below two scenarios.

1. A fixed amount is added based on the price category of the original book.
2. An additional amount of Rs.10 is added for each day after the due date.

The fixed amount of penalty is calculated based on the below categories.

Price Category	Fixed Fine Amount
1. Rs.100 - Rs.500	Rs.50
2. Rs.600 - Rs.1000	Rs.150
3. More than Rs.1000	Rs.250

Program should ask the librarian to enter the corresponding membership id, member's name, number of days late and price category to which the book belongs to and display the total fine as follows. (Hint : If a member has a 5 days due for the book "Divergent" he/she should be charged with 250 + 5*10 = 300)

```
Example output:
Enter Your Choice: 5
                Fine Details
-----------------------------------------------------------
Price Range.             Fine
------------------------------------------------------------
1.Rs.100 - Rs.500        Rs.50
2.Rs.600 - Rs.1000       Rs.150
3.More than Rs.1000      Rs.250

Enter membership id: 10054
Enter member_name: Bindam
Enter number of days late: 5
Select price range: 3
The total fine is: 300.00
```

PART B - (PYTHON PROGRAMMING)

Create a folder named CSC1041_End_S20XXX in your H drive. Copy the shared text files into the newly created folder. Save your Python programs inside the same folder as S20XXXQ2.py, S20XXXQ3.py, S20XXXQ4.py for each question where S20XXX is your registration number.

2. [30 Marks]
Write a python programme to calculate and output final GPAs.

a) Create a dictionary to store the following course details of a student.

Course code	Course name	No.of credits	Assignment marks	Exam marks	Labsheet marks
CS104	C Programming	1	80,50,40,20	75,75	78.20, 77.20
ST102	Probability Theory	3	82,56,44,30	80,80	67.90, 78.72
MT101	Abstract Algebra	2	77,82,23,39	78,77	80, 80
CS101	Introduction to Computer Science	3	67,55,77,21	40,50	69, 44.56
MT105	Real Analysis	3	29,89,60,56	65,56	50, 40.6

b) Define a generic function to calculate the **total average marks** for a given list of values.

c) Using the function created in (b), define a function to calculate the final mark for each subject according to the below criteria
- Assignments 10%
- Exams 70%
- Labsheets 20%

d) Define a function to calculate the letter grade of each subject.

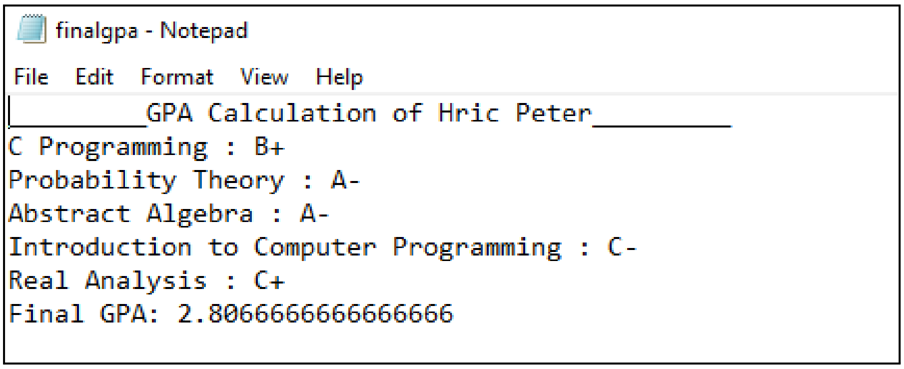
- Marks > 90 ⇒ 'A+'
- Marks > 80 ⇒ 'A'
- Marks > 75 ⇒ 'A-'
- Marks > 70 ⇒ 'B+'
- Marks > 65 ⇒ 'B'
- Marks > 60 ⇒ 'B-'
- Marks > 55 ⇒ 'C+'
- Marks > 50 ⇒ 'C'
- Marks > 45 ⇒ 'C-'
- Marks > 40 ⇒ 'D+'
- Marks > 30 ⇒ 'D'
- Marks < 30 ⇒ 'E'

e) Open the text file named **gradepoints.txt** in your H drive. That file contains the points for each letter grade. Store that data using a suitable python collection data type.
f) Calculate the letter grades for each course and print the final marks, letter grade for each course.

g) Define a function to calculate GPA. [**Hint:** GPA = ∑ (Points for letter grade * No.of credits) / Total no. of credits]

h) Print the final GPA point for the student.

i) Update the "finalgpa.txt" text file with the course results and the final gpa.
Output:

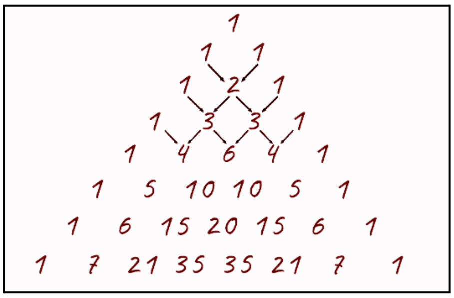


```
finalgpa - Notepad
File  Edit  Format  View  Help
_________GPA Calculation of Hric Peter_________
C Programming : B+
Probability Theory : A-
Abstract Algebra : A-
Introduction to Computer Programming : C-
Real Analysis : C+
Final GPA: 2.8066666666666666
```

j) Name of the student is mistakenly printed as *Hric* in the text file. The correct name of the student is *Eric Peter*. Create a function called **HtoE()** in order to print the full content of the file by replacing the letter H to E.

3. [12 Marks]

Pascal Triangle is a number pattern in the shape of a triangle. Each number in the triangle is the sum of the two numbers above it. Create a function to print Pascal Triangle. Take the number of rows as user input.



4. [08 Marks]

Create a python program to check the validity of a password. The password should contain following conditions:

1. Minimum 8 characters
2. The alphabet must be between a-z
3. At Least one should be of upper case
4. At Least one number between 0-9
5. At Least one special character from **[$, % , & , _]**

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